

PROJECT DOCUMENTATION

HackMatrix 2026 · Round 2

1. TEAM INFORMATION

Team Name	MINDMATES
Team Leader	Uddish Agarwal
Contact	uddishagarwal81@gmail.com
Event	HackMatrix 2026 — Round 2
Institute	Madhav Institute of Technology & Science, Gwalior (M.P.)

2. LINKS

GitHub Repository (public)
<https://github.com/Uddish12dev/Traffic-Light-Sync-AI---HackMatrix2026>
Demo Video Link
https://drive.google.com/file/d/167MKkDB4uFF3K7usSJVGOAYA_dCOefwp/view?usp=sharing

3. PROJECT NAME / TITLE

Signal Sync AI — AI-Powered Adaptive Traffic Signal Controller

4. PLATFORM PREVIEW

Signal Sync AI is a Python-based simulation platform that models a multi-intersection traffic network. A web-ready dashboard (CLI + optional Flask UI) displays live signal states, predicted waiting times, and an event log for every intersection in real time. The architectural schematic below summarises the data flow:

Traffic Sensors	→	Data Preprocessing	→	AI Model Inference
		(count · speed · occ · flow · time)		(Random Forest)
				↓
			Safety Conflict-Map Check	
				↓
			Signal Actuation (Green / Yellow / Red)	

5. SUMMARY

Signal Sync AI replaces traditional fixed-timer traffic signals with a real-time, AI-driven controller that predicts vehicle waiting times and recommends the optimal signal state (Red / Yellow / Green) at each intersection. A hard-coded conflict-map safety layer guarantees that no two conflicting directions can ever both be Green simultaneously. Every sensor reading and decision is logged automatically, enabling continuous model improvement. The system is built for multi-intersection networks and is designed to scale from a hackathon prototype to a city-wide smart traffic deployment.

6. PROBLEM BEING SOLVED

- Traditional signals run on fixed timers — they do not react to real-time traffic density.
- Peak-hour congestion causes unnecessary waiting, wasted fuel, and higher emissions.
- No existing low-cost system provides AI-driven, safety-verified adaptive switching for multiple intersections simultaneously.
- Longer ambulance / emergency vehicle response times due to unresponsive signals.

7. USP — UNIQUE SELLING POINT

USP	Detail
Dual-output AI model	Single inference gives both expected waiting time AND the recommended signal colour
Safety-first architecture	Rule-based conflict-map runs BEFORE any green is issued — AI can never override safety
Primary / Secondary roles	Only the 2 relevant intersections query the AI at any moment — O(1) decisions regardless of network size
Self-improving loop	Every decision is logged; dataset grows and model can be retrained incrementally
Zero hardware dependency	Runs on pure sensor data; deployable on a Raspberry Pi or in the cloud

8. KEY FEATURES

- **Real-time adaptive control** — Adjusts signal timing based on live sensor data (vehicle count, speed, occupancy, flow rate, time of day).
- **Dual AI predictions** — One model simultaneously predicts waiting time (regression) and optimal signal colour (classification).
- **Intersection role management** — Assigns Primary (currently Green) and Secondary (currently Red) roles dynamically.
- **Safety conflict-map** — Rule-based hard override prevents any conflicting green — no exceptions.
- **Continuous feedback loop** — All decisions and sensor readings are logged, enabling incremental model retraining.
- **Scalable design** — Architecture supports N intersections with O(1) AI inference at each step.
- **Simulation-ready** — Standalone Python controller for demonstration and testing without physical hardware.

9. TECH STACK

Layer	Technology	Purpose
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AI / ML	scikit-learn	Random Forest Classifier (signal colour) + Regressor (waiting time)
Language	Python 3.11	Core simulation, controller logic, and model inference
Data	Pandas / NumPy	Sensor data ingestion, feature engineering, and logging
Visualisation	Matplotlib (optional)	Real-time signal state charts and performance metrics
Deployment	Local / Vercel / Pi	CLI prototype; Flask API for web dashboard; Raspberry Pi for edge
Version Control	Git + GitHub	Public repository with full commit history

10. FUTURE SCOPE (OPTIONAL)

- **Real sensor integration** — Connect to inductive loop sensors, cameras, or IoT nodes replacing simulated data.
- **Emergency vehicle override** — Detect emergency signals and automatically clear the path across intersections.
- **Deep reinforcement learning** — Replace Random Forest with a DRL agent (PPO / DQN) trained on real-world data for city-scale optimisation.
- **V2I communication** — Vehicle-to-Infrastructure handshake so connected cars report their speed and position directly.
- **City-wide dashboard** — React/Next.js web app with live map overlay showing intersection states, wait-time heatmaps, and emission estimates.
- **Multi-modal traffic support** — Extend model to handle pedestrian crossings, cyclists, and bus priority lanes.

Document prepared for HackMatrix 2026 · Round 2 — Team MINDMATES — MITS Gwalior